

2.4 Integrating by Partial Fractions

Motivating Examples:

Suppose we wish to find $\int \frac{5x^2 - x - 1}{2x^3 - x^2 - x} dx$. How could one do this?

One can see that a u -substitution has the potential to be quite messy. One also might wonder: “Maybe we can split the fraction up and create several simpler integrals.” This second idea turns out to work beautifully in this example (if one knows how to split the fraction up properly):

$$\int \frac{5x^2 - x - 1}{2x^3 - x^2 - x} dx = \int \left(\frac{1}{x} + \frac{1}{2x+1} + \frac{1}{x-1} \right) dx = \int \frac{1}{x} dx + \int \frac{1}{2x+1} dx + \int \frac{1}{x-1} dx = \ln|x| + \frac{1}{2} \ln|2x+1| + \ln|x-1| + C.$$

So, a good question to ask from here is: Is there a nice algorithm for splitting a rational function up in order to make it easy to integrate?

Now, suppose we wish to find: $\int \frac{x^3 + x}{x-1} dx$. How could one do this?

In this case since the degree of the polynomial in the numerator is greater than the degree of the polynomial in the denominator, polynomial long division is a good idea. In particular when we perform the long division we obtain:

$$\begin{array}{r} x^2 + x + 2 \\ x-1 \overline{) x^3 + 0x^2 + x + 0} \\ \underline{-(x^3 - x^2)} \\ x^2 + x + 0 \\ \underline{-(x^2 - x)} \\ 2x + 0 \\ \underline{-(2x - 2)} \\ 2 \end{array}$$

Now, the integral becomes:

$$\int \frac{x^3 + x}{x-1} dx = \int \left(x^2 + x + 2 + \frac{2}{x-1} \right) dx = \frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x + 2\ln|x-1| + C.$$

Note: We will see in this section that it is generally a good idea to use polynomial long division when we are integrating a rational function in which the degree of the numerator is greater than or equal to the degree of the denominator (such rational functions are called improper and rational functions in which the degree of the numerator is less than the degree of the denominator are called proper).

- Outline:** In this section our general outline for integrating will be to:
1. Perform polynomial long division if the rational function is improper.
 2. Express any proper rational functions as a sum of its *partial fractions* where the partial fractions are of the form: $\frac{A}{(ax+b)^i}$ or $\frac{Ax+B}{(ax^2+bx+c)^j}$.

Note: Similar to the text we will consider four separate cases.

Case 1 (The denominator is the product of distinct linear factors):

Example 1: Find the exact value of: $\int_0^1 \frac{x-4}{x^2-5x+6} dx$.

First, we note that we are integrating a proper rational function. This means we will not use polynomial long division. This brings us to step 2 of our outline. We wish to find A and B so that:

$$\frac{x-4}{x^2-5x+6} = \frac{x-4}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2}.$$

Now, we manipulate this equation to find A and B :

$$\begin{aligned} \frac{x-4}{(x-3)(x-2)} &= \frac{A}{x-3} + \frac{B}{x-2} \\ \Rightarrow x-4 &= A(x-2) + B(x-3) \\ \Leftrightarrow x-4 &= (A+B)x - 2A - 3B \\ \Rightarrow \begin{cases} A+B=1 \\ -2A-3B=-4 \end{cases} \end{aligned}$$

After we solve this system we see $A=-1$ and $B=2$.

Note: There is a much slicker way to solve for A and B . In particular when we get $x-4 = A(x-2) + B(x-3)$ we know that this equation needs to hold for all values of x . In particular it should hold when $x=2$ and when $x=3$. If we plug two in for x the equation becomes $2-4 = A(2-2) + B(2-3) \Leftrightarrow -2 = -B \Leftrightarrow 2 = B$ and if we plug three in for x the equation becomes: $3-4 = A(3-2) + B(3-3) \Leftrightarrow -1 = A$.

This means that:

$$\begin{aligned} \int_0^1 \frac{x-4}{x^2-5x+6} dx &= \int_0^1 \left(\frac{-1}{x-3} + \frac{2}{x-2} \right) dx = \left(-\ln|x-3| + 2\ln|x-2| \right) \Big|_0^1 \\ &= -\ln(2) + 2\ln(1) + \ln(3) - 2\ln(2) = \ln(3) - 3\ln(2) = \ln(3) - \ln(8) = \ln\left(\frac{3}{8}\right). \end{aligned}$$

Example 2 (you try): Find $\int \frac{x^2 + 2x - 1}{x^3 - x} dx$.

First, we note that we are integrating a proper rational function. This means we will not use polynomial long division. This brings us to step 2 of our outline. We wish to find A , B , and C so that:

$$\frac{x^2 + 2x - 1}{x^3 - x} = \frac{x^2 + 2x - 1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}.$$

Now, solve for A , B , and C by using the “slick” method mentioned above:

$$\frac{x^2 + 2x - 1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} \Rightarrow x^2 + 2x - 1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1).$$

If we let $x = 1$ we see that: $2 = 2B \Leftrightarrow B = 1$. If we let $x = 0$ we see that: $-1 = -A \Leftrightarrow A = 1$. If we let $x = -1$ we see that: $-2 = 2C \Leftrightarrow C = -1$.

This means that:
$$\int \frac{x^2 + 2x - 1}{x^3 - x} dx = \int \left(\frac{1}{x} + \frac{1}{x-1} + \frac{-1}{x+1} \right) dx = \ln|x| + \ln|x-1| - \ln|x+1| + C.$$

Case 2 (the denominator is a product of linear factors, some of which are repeated)

In this case if $(ax + b)^n$ (where b is a real number and $a \neq 0$) is a factor of the denominator, then the partial fractions corresponding to this factor are: $\sum_{i=1}^n \frac{A_i}{(ax + b)^i}$.

We will illustrate this with an example.

Example 3: Find $\int \frac{x^2 - 5x + 16}{(2x+1)(x-2)^2} dx$.

We first note that there is no need for polynomial long division. Now, we wish to find A , B , and C so that they

satisfy:
$$\frac{x^2 - 5x + 16}{(2x+1)(x-2)^2} = \frac{A}{2x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}.$$

Note how the number of partial fractions is the same as the number of linear factors of the denominator. This holds when all the factors of the denominator are linear.

Now, we solve for A , B , and C :
$$\frac{x^2 - 5x + 16}{(2x+1)(x-2)^2} = \frac{A}{2x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$
$$\Rightarrow x^2 - 5x + 16 = A(x-2)^2 + B(x-2)(2x+1) + C(2x+1).$$

If we let $x = 2$ we see that: $10 = 5C \Leftrightarrow C = 2$. If we let $x = -\frac{1}{2}$ we see that: $\frac{75}{4} = \frac{25}{4}A \Leftrightarrow A = 3$. We cannot make A and C disappear with a particular value of x . So, we let $x = 0$ and we use what we know to obtain: $16 = 4A - 2B + C \Leftrightarrow 16 = 4(3) - 2B + 2 \Leftrightarrow 2 = -2B \Leftrightarrow B = -1$.

This means that:

$$\int \frac{x^2 - 5x + 16}{(2x+1)(x-2)^2} dx = \int \left(\frac{3}{2x+1} + \frac{-1}{x-2} + \frac{2}{(x-2)^2} \right) dx = \frac{3}{2} \ln|2x+1| - \ln|x-2| + \frac{-2}{x-2} + C .$$

Note: If you can't do the last integration step in your head you can always split the integral up and make u substitutions when you need them.

Case 3 (the denominator contains irreducible quadratic factors none of which is repeated):

The idea is that if $ax^2 + bx + c$ is an irreducible quadratic factor (meaning that $b^2 - 4ac < 0$) appearing in the denominator of a rational function, the partial fraction corresponding to this factor is of the form: $\frac{Ax + B}{ax^2 + bx + c}$.

We illustrate this with an example.

Example 4: Find $\int \frac{4x^2 + 2}{x^3 + x} dx$.

We first note that there is no need for polynomial long division. Then, following the above idea we want to find A , B , and C so that $\frac{4x^2 + 2}{x^3 + x} = \frac{4x^2 + 2}{x(x^2 + 1)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x}$.

Note: the quadratic factor $x^2 + 1$ is an irreducible quadratic factor.

Now, we solve for A , B , and C . We note that: $\frac{4x^2 + 2}{x(x^2 + 1)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x} \Rightarrow 4x^2 + 2 = (Ax + B)x + C(x^2 + 1)$. If

we let $x = 0$ we obtain $2 = C$. Now,

$$4x^2 + 2 = (Ax + B)x + C(x^2 + 1) \Leftrightarrow 4x^2 + 2 = (Ax + B)x + 2(x^2 + 1) \Leftrightarrow 2x^2 = Ax^2 + Bx .$$

We conclude that $A = 2$ and $B = 0$.

Thus, the integral becomes: $\int \frac{4x^2 + 2}{x^3 + x} dx = \int \left(\frac{2x}{x^2 + 1} + \frac{2}{x} \right) dx = \int \frac{2x}{x^2 + 1} dx + \int \frac{2}{x} dx$. Now, we have two indefinite integrals to find. For the first of these indefinite integrals we make the substitution:

$$u = x^2 + 1 \Rightarrow du = 2x dx . \text{ Now, we calculate: } \int \frac{2x}{x^2 + 1} dx + \int \frac{2}{x} dx = \int \frac{1}{u} du + 2 \ln|x| + C \\ = \ln|u| + 2 \ln|x| + C = \ln|x^2 + 1| + 2 \ln|x| + C .$$

Example 5 (you try):

(a) Find $\int \frac{2x^2 - x + 4}{x^3 + 4x} dx$. *Hint: Recall that $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$ where $a \neq 0$.*

We note that $\int \frac{2x^2 - x + 4}{x^3 + 4x} dx = \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx$. Since $x^2 + 4$ is an irreducible quadratic, we know we are in

case 3. So, we seek A , B , and C so that

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4} \Rightarrow 2x^2 - x + 4 = A(x^2 + 4) + (Bx + C)x \Rightarrow 2x^2 - x + 4 = (A + B)x^2 + Cx + 4A. \text{ We}$$

immediately see from this that: $C = -1$ and $4A = 4 \Rightarrow A = 1$. Then, $A + B = 2 \Rightarrow B = 1$. So, the integral becomes:

$$\int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx = \int \left(\frac{1}{x} + \frac{x-1}{x^2 + 4} \right) dx = \int \left(\frac{1}{x} + \frac{x}{x^2 + 4} - \frac{1}{x^2 + 4} \right) dx = \ln|x| + \frac{1}{2} \ln(x^2 + 4) - \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C.$$

(b) Find $\int \frac{x-1}{4x^2 - 4x + 3} dx$. *Hint: The quadratic on the bottom is irreducible. So, you will not be splitting this rational function up. Try to complete the square on the bottom of this fraction.*

Solution: We note $4x^2 - 4x + 3 = 4(x^2 - x) + 3 = 4\left(x^2 - x + \frac{1}{4}\right) - 1 + 3 = 2^2\left(x - \frac{1}{2}\right)^2 + 2$. Now, we

have: $\int \frac{x-1}{4x^2 - 4x + 3} dx = \int \frac{x-1}{(2x-1)^2 + 2} dx$. We make the substitution $u = 2x - 1 \Rightarrow du = 2dx \Leftrightarrow \frac{1}{2} du = dx$.

We note also that $u = 2x - 1 \Leftrightarrow \frac{u+1}{2} = x$. Now, the integral becomes

$$\begin{aligned} \int \frac{x-1}{(2x-1)^2 + 3} dx &= \frac{1}{2} \int \frac{\frac{u+1}{2} - 1}{u^2 + 2} du = \frac{1}{2} \int \frac{\frac{1}{2}(u-1)}{u^2 + 2} du = \frac{1}{4} \int \frac{u-1}{u^2 + 2} du \\ &= \frac{1}{4} \left[\int \frac{u}{u^2 + 2} du - \int \frac{1}{u^2 + 2} \right] = \frac{1}{4} \left[\int \frac{u}{u^2 + 2} du - \int \frac{1}{u^2 + \sqrt{2}^2} \right]. \end{aligned}$$

Now, we use the hint from (a) to finish the problem:

$$\begin{aligned} \frac{1}{4} \left[\int \frac{u}{u^2 + 2} du - \int \frac{1}{u^2 + \sqrt{2}^2} \right] &= \frac{1}{4} \left[\frac{1}{2} \ln(u^2 + 2) - \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) \right] + C \\ &= \frac{1}{8} \ln((2x-1)^2 + 2) - \frac{1}{4\sqrt{2}} \tan^{-1}\left(\frac{2x-1}{\sqrt{2}}\right) + C. \end{aligned}$$

Note: part (b) is important because it shows a strategy that will always allow us to integrate an expression of the form $\frac{Ax+B}{ax^2+bx+c}$ where the quadratic is irreducible.

Case 4 (the denominator contains a repeated irreducible quadratic factor)

Case 4 generalizes Case 3 similar to the way in which Case 2 generalizes Case 1. Specifically if the denominator of a rational function contains a repeated irreducible quadratic factor: $(ax^2 + bx + c)^r$, then the

partial fractions corresponding to this factor are: $\sum_{i=1}^r \left(\frac{A_i x + B_i}{(ax^2 + bx + c)^i} \right)$.

Example 6: Find a partial fraction decomposition of: $\frac{x^2 + x + 1}{(x^2 + 1)^2}$.

We wish to find A , B , C , and D so that $\frac{x^2 + x + 1}{(x^2 + 1)^2} = \frac{Ax + B}{(x^2 + 1)} + \frac{Cx + D}{(x^2 + 1)^2}$. Now we use this equation to find A , B ,

$$\frac{x^2 + x + 1}{(x^2 + 1)^2} = \frac{Ax + B}{(x^2 + 1)} + \frac{Cx + D}{(x^2 + 1)^2} \Rightarrow x^2 + x + 1 = (Ax + B)(x^2 + 1) + Cx + D$$

C , and D : $x^2 + x + 1 = Ax^3 + Bx^2 + Ax + B + Cx + D \Leftrightarrow 0x^3 + x^2 + x + 1 = Ax^3 + Bx^2 + (A + C)x + B + D$ From the

$$\Leftrightarrow \begin{cases} A = 0 \\ B = 1 \\ A + C = 1 \\ B + D = 1 \end{cases}$$

above system we conclude: $A = 0$, $B = 1$, $C = 1$, and $D = 0$. Thus, we have: $\frac{x^2 + x + 1}{(x^2 + 1)^2} = \frac{1}{(x^2 + 1)} + \frac{x}{(x^2 + 1)^2}$.

Note: One can easily integrate the expression we obtained on the right hand side.